

Toward Real-Time Volcano Seismic Events' Classification: A New Approach Using Mathematical Morphology and Similarity Criteria

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Abstract—This work proposes a new approach based on a suit combination of mathematical morphology and similarity criteria techniques to classify long-period and volcano-tectonic seismic events of the Cotopaxi volcano. The proposed method explores the seismic signal domain to compute a new feature space based on the edges map of the seismic events pattern represented in the gray-level spectrogram images, which is used to feed a set of similarity-based classifiers. The L_2 -norm was selected as the best metric to be implemented by the proposed method. In terms of classification performance, the L_2 -norm was statistically superior in the D1 data set (seismic events with overlapped signals of nonvolcanic origin) and similar in the D2 data set (events without overlapped signals) with respect to the other metrics, reaching accuracy mean scores of 93.34% and 96.88%, respectively. These results demonstrated that the computed edges map feature space is a better environment for separating both seismic events compared with the original gray-level space. Regarding the execution time, total time (TT) and time per-sample (TS) did not exceed 0.388 and 0.002 s during the training stage, respectively. During the testing stage, a TS of no more than 0.012 s was achieved. Finally, its execution time is faster, and the algorithm complexity is lower compared with the state-of-the-art methods, which makes it a practical and beneficial scheme to implement for real-time seismic events' classification.

Index Terms—Distance metrics, mathematical morphology-based features, seismic events' classification, seismic pattern generation, similarity criteria.

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I. INTRODUCTION

ERUPTING volcanoes pose a degree of risk to people and infrastructure [1]. Although volcanic eruptions are unpredictable, volcanic observatories use seismic monitoring techniques, among other technologies, to estimate when they are likely to happen once an unrest status has been detected [2]. In such a sense, early warning systems are related to guarantee fast actions in case of an eruption, including opportune alarms, rapid generation of high-quality information about the situation, and efficient resource allocation to keep people and communities safe. An example of such a system is the National Volcano Early Warning System (NVEWS) by the U.S. Geological Survey (USGS) Volcano Hazards Program [3]. In this context, real-time monitoring of critical parameters then becomes essential [4]. Nevertheless, the real-time term is a physic magnitude that is impossible to reach due to relativity. Despite that, a real-time system can be described as one that could inter-react and control the environment by receiving data, processing them, and returning the results quickly enough to modify the environment [5].

In volcanology, some processes require large amounts of data to assess the situation; therefore, we can talk of real-time if the seismic events are detected and classified continuously as they occur. However, classification depends on feature spaces and machine learning classifiers (MLCs), which, usually, are combined to fulfill the task after the seismic data samples are collected.

Regarding the feature space, different features have been computed from raw seismogram recordings to encode different types of seismic events, involving features from time [6], [7], frequency [8], and scale domains (i.e. wavelet-based features) [9], and also from Mel frequency cepstral coefficients [10], discrete cosine transform [11], linear predictive coding coefficients [12], and, recently, the intensity statistics, shape, and texture of the spectrogram image [13]. Mathematical morphology has also been used in seismic signal analysis but mostly for denoising purposes, for example, to suppress both incoherent [14] and coherent [15] noise and to unveil weak signals from extremely noisy observed data [16].

Similarly, for the classification, several MLCs have solved the volcano seismic events' classification problem from different learning perspectives, ranging from supervised techniques, such as artificial neural networks (ANNs) [17]–[19], deep neural networks [20]–[22], support vector machine (SVM) [8], [12], [23], random forest [13], [24], decision trees (DTs) [25], [26], the hidden Markov model (HMM) [27], [28], evolutionary algorithms [29], [30], and Gaussian mixture models (GMM) [31] to other techniques using unsupervised [32], [33] and semisupervised learning [9]. Nevertheless, the abovementioned approaches incur a high algorithm complexity due to the feature calculation, selection, and classification steps required to carry out the solutions, which is impractical for real-time scenarios. As a result of this constraint, volcano observatory stations still advocate using manual classification by specialists over automatic classification methods even though the manual classification of seismic events could lead to delays and errors due to human subjectivity.

On the other hand, the similarity criterion is a popular concept related to content-based image retrieval systems, which can be used as an alternative for classification. Classification using similarity functions, usually distance functions, efficiently measures (rank) the matching of a query image versus a commonly large-scale database [34]–[36]. However, currently, there is little evidence in the literature about using this concept for classifying volcanic activity. For example, the use of spectrogram cross correlation as a similarity measure to classify seismic signals from the Llaima volcano (Chile) was proposed in [37].

Against this background, we think that incorporating similarity criterion techniques in the volcano seismic event classification context could optimize the processing time by decreasing the algorithmic complexity while retaining satisfactory classification performance. Thus, in this article, we propose an innovative method based on a suit combination of mathematical morphology and similarity criteria techniques to classify long-period (LP) and volcano-tectonic (VT) seismic events in the Cotopaxi volcano.

The contribution behind this proposal is related to the following.

- 1) *Creating a New Feature Space*: We used a morphological gradient operation to calculate the edges map of the seismic event pattern represented in the gray-level spectrogram images. Therefore, our feature space contains only information about the contours of the pattern.
- 2) *Using Similarity Criteria as a Classifier*: Since the feature space is a vector space, the traditional MLC was replaced by a distance function that measures the closeness of two feature vectors and determines the output class label.
- 3) *Faster-Performing Classification Model*: Calculation of distances (similarity) between feature vectors is performed in linear time. Therefore, time execution (faster) and complexity (lower) are beneficial for implementing real-time volcano monitoring compared with the state-of-the-art methods.

The rest of this article is organized as follows. Section II briefly describes previously developed approaches in the

context of volcano seismic events' classification. Section III presents a detailed description of the proposed method and the experimental setup designed for its evaluation. Section IV outlines the obtained results from the classification of seismic events on an experimental data set, an exploratory comparison of the original versus computed feature space, and the proposed method against previously developed approaches available in the literature. Section V summarizes the principal limitations of the proposed method. Finally, Section VI remarks on the primary contributions and future work.

II. RELATED WORK

During the last decade, many successful approaches were reported in the literature for volcano seismic events' classification. They often employ signal processing techniques in conjunction with machine learning schemes to achieve highly accurate event classifiers aimed at monitoring active volcanoes [38]. For example, Lara-Cueva *et al.* [39] reported a performance comparison between three well-known supervised MLCs, the k-nearest neighbor algorithm (kNN), DT, and an ANN, where the kNN reached the highest accuracy (ACC) value of 98%. Benitez *et al.* [40] presented a recognition system based on HMM methods to classify VT earthquakes, LP, volcanic tremor, and hybrid events. The system was tested in a database, containing feature vectors of computed log-energies. The reported ACC score on the experimental database was 90%. Also, their application in another database provided an ACC score of 95%.

In [31], extensive experimentation using five filter-based feature selection methods in combination with a Gaussian mixture classifier was made. The reported method was able to determine the best classification scheme for the classification of LP events, VT events, and both events together (as disjoint classes), yielding ACC scores of 95.62%, 96.71%, and 96.70%, respectively. While a computer-based classifier to automatically identify four seismic event classes of the Llaima volcano was developed in [23], it used a set of computed features from the amplitude, frequency, and phase of the seismic signals to form training and test partitions to feed support vector machine (SVM) models. The reported performance indicated good recognition results for the Llaima volcano, yielding a classification ACC score above 80%.

Lately, there is a tendency to classify seismic events based on their spectrogram images using shallow and deep learning techniques. For example, in a previous work [13], we developed a volcano seismic signal descriptor to classify LP and VT seismic events. The method computes a set of features based on the intensity statistic of pixels, shape, and texture of segmented seismic patterns in the gray-level spectrogram images. Subsequently, five MLCs with different taxonomies were fed with the computed feature vectors to determine the effectiveness of the formed feature space. Among all MLCs, the feed-forward backpropagation (FFBP) ANNs classifier demonstrated to be the best classification model, reaching area under the curve of the receiver operating characteristic (AUC) values of 0.95 and 0.96, in a data set with and without events overlapping, respectively. These features were benchmarked

with others from the time, frequency, and scale domains in [7], highlighting their superiority.

In [17], three different convolutional neural network (CNN) architectures were built to classify spectrogram images of LP and VT seismic events. The best AUC score of 0.94 and the ACC value of 94.13% were reached by a CNN model with a (3×3) convolutional and pool kernel size and 1×10^{-3} as learning rate on a data set of 668 samples. On the other hand, a deep neural network to discriminate seismic activities by analyzing the spectrogram shapes was developed in [22], reaching outstanding results with ACC rates higher than 95%. Similarly, in [20], six deep learning models based on CNN and recurrent neural networks (RNN) were proposed to differentiate between two classes of seismic events. The classifiers were validated in a data set of 668 spectrogram images of LP and VT events. The best classifier was formed by two convolutional layers, which attained AUC and ACC scores of 0.98 and 95%, respectively. Moreover, Titos *et al.* [41] proposed several RNN-based architectures beyond classical RNNs, such as long short-term memory (LSTM) and gated recurrent unit (GRU) to detect and classify continuous sequences of volcano-seismic events at the Deception Island Volcano, attaining close to 94% of events correctly detected and classified. A similar performance has been achieved by CNNs [42] and Bayesian neural networks [43] based on transfer learning schemes.

On the other hand, unsupervised learning approaches focusing on volcanoes and their seismic activities have been less explored, but self-organizing map (SOM) models seem to be the most popular. In [32], an SOM model focusing on volcanic wavefield patterns was used to analyze the seismicity of Mount Merapi (Indonesia). They obtained errors of 6% and 26% for VT and rockfall event classification, respectively. However, when both events were combined into one cluster class, the error was significantly reduced to 12%. Similarly, in [44], SOM and k-means models were used to classify volcanic signals recorded from the Tungurahua volcano (Ecuador), attaining ACC values of 91% and 86% for noise and infrasound signals, respectively. While, in [45], SOM and clustering-based models were integrated to build the “KKAnalysis” software tool that takes less than a minute to classify volcanic tremor data related to Mount Etna (Italy), reaching an ACC value of 90%.

Furthermore, in [33], an exploration of six different clustering-based classifiers was made, including the k-means, BFR, CURE, BIRCH, expectation–maximization, and spectral-clustering methods. These classifiers were validated on a data set containing LP and VT seismic events and some events with overlapped signals on nonvolcanic origins, such as rockfalls or icequakes that were recorded at the same time as LP or VT events occurrence. The best classification result was reached by the BIRCH method with $k = 2$, touching an ACC score of 88%. The spectral-clustering with $k = 3$ method was the only one who successfully detected seismic events with an overlapped signal, attaining a true positive rate score of 0.94.

Mathematical morphological filtering (MMF) has been initially proposed to process seismic data by Wang *et al.* [46]. For instance, Huang *et al.* [16] describe a method to

successfully detect extremely weak microseismic signals for high-resolution hydraulic fracturing monitoring. Concretely, the technique decomposes a noisy signal into several morphological multiscale components. Moreover, in [14], a method for denoising seismic data based on planar mathematical morphological filtering (PMMF) is proposed. They found that PMMF can significantly suppress noise because it barely overlaps the seismic signal in morphological scale on a plane. In a similar work [15], MMF was employed to coherent noise attenuation on seismic signals. They showed that their approach is more robust than mean and median filters in removing coherent noise. Nonetheless, mathematical morphology has not yet been used in the context of volcanic seismic events' classification.

In [37], spectrogram cross correlation (SPCC) was used as a measure of similarity between the shape of spectrograms of seismic events. The SPCC was used along with a KNN algorithm to classify seismic signals without requiring a pre-processing step centered on the feature analysis. A sensitivity of over 95% was achieved for three out of the five-event classes considered on signals collected from the Llaima volcano.

Given the background from this section, it is possible to notice that most previously developed methods for volcano seismic events' classification are based on advanced MLCs, which demands plenty of time in the training stage. Therefore, we expect that the contributions of the proposed method address the existing limitations of building faster models with lower algorithmic complexity and without losing classification performance.

III. MATERIALS AND METHODS

A. Experimental Data Sets

This work considered the use of the publicly available MicSigV1 data set from the ESeismic,¹ where you should register and fill a disclaimer agreement in order to obtain the data from the repository [47]. ESeismic is the first annotated Ecuadorian volcano seismic repository offering samples of different types of events, such as LP, VT, regional (REG), hybrid (HB), and icequake (ICE) from Cotopaxi volcano. Moreover, since this data set comes from a real-world scenario, some samples also contain overlapped signals (i.e., signals that could occur at the same time or immediately after the actual event).

We created two experimental data sets, in which most of the events are subsets of the MicSigV1, to feed the proposed method. The D1 data set contains 668 raw signals of LP (587 instances) and VT (81 instances) event classes, including some samples of both classes with overlapped signals (65 instances). On the other hand, the D2 data set is a subset of the D1 data set (excluding samples with overlapped signals) containing 603 raw signals of LP (538 instances) and VT (65 instances) event classes. Fig. 1 shows an example of the seismic events in these data sets. It should be noted that, since recorded signals are dependents on the seismic station location [47], all samples on both data sets correspond to a

¹ESeismic repository was provided by courtesy of the Instituto Geofísico of Escuela Politécnica Nacional (IGEPN) and collaborators, and it is available at http://www.igepn.edu.ec/eseismic_web_site/index.php

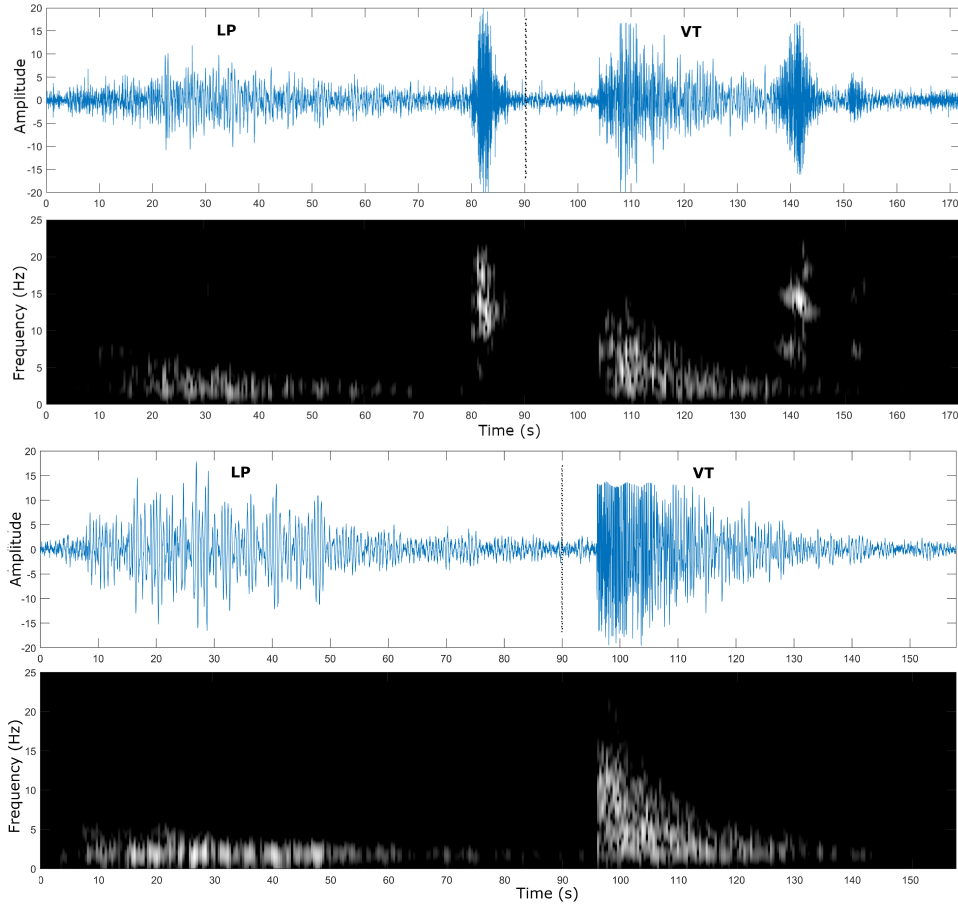


Fig. 1. Examples of signals in the time domain and their respective filtered spectrograms corresponding to LP and VT events (top) with and (bottom) without overlapped signals taken from the MicSigV1 data set. The time signals were normalized by their maximum absolute value.

single station deployed on the volcano's flanks to avoid mixing signals from different locations within the seismic network. Also, seismic events, such as REG (27 instances), HB (eight instances), and ICE (seven instances), were not considered due to their small representation in the MicSigV1 data set. These data sets will help us to gather information about the performance of the proposed method.

B. Proposed Method

The proposed method combines the mathematical morphology to compute the edges map-based features of the seismic event pattern formed in the gray-level spectrogram images and a distance metric as similarity criteria to classify both the LP and VT seismic events. Fig. 2 shows the entire workflow of the proposed method for the training and test stages. A detailed description is presented next:

1) *Spectrogram Generation*: The short-time Fourier transform (STFT) [48] was used to generate the gray-level spectrogram image for each signal on the experimental data sets. The transformation used a Kaiser window [49] with 1.5 s sampling window size and 75% of overlapping to conserve the boundary information of the window and, thus, to maximize the energy in the main part of the signal. Then, each spectrogram image was filtered using a power threshold of 0 dB (experimentally

determined) to remove frequency components with low magnitude while retaining those related to the seismic event (see steps 1 and 2 in Fig. 2).

2) *Events Atlas Calculation*: The atlas of seismic events is the most common representation of the event's pattern among all spectrogram images. Thus, we average all the samples of LP and VT spectrogram images ($\mathbf{I}^{LP}$ and $\mathbf{I}^{VT}$), respectively, to produce their corresponding atlas ($\mathbf{A}^{LP}$ and $\mathbf{A}^{VT}$), which is the mean image of the events given by

$$\mathbf{A}^{LP} = \frac{\sum_{i=1}^j \mathbf{I}^{LP}_i}{j} \quad \mathbf{A}^{VT} = \frac{\sum_{i=1}^k \mathbf{I}^{VT}_i}{k}$$

where j and k are the number of $\mathbf{I}^{LP}$ and $\mathbf{I}^{VT}$, available in the training set, respectively. An example of the atlas calculation is shown in step 3 in Fig. 2.

3) *Feature Vector Calculation*: Feature extraction is about obtaining as much information as possible from the spectrogram image under study. In this way, each sample remains represented (encoded) by some features that will help to index it anytime.

Our feature vectors contain information about the edges map of the seismic event pattern formed in the gray-level spectrogram images. This information was computed using a morphological gradient operation, a mathematical morphology technique used in digital image processing [50]. This technique

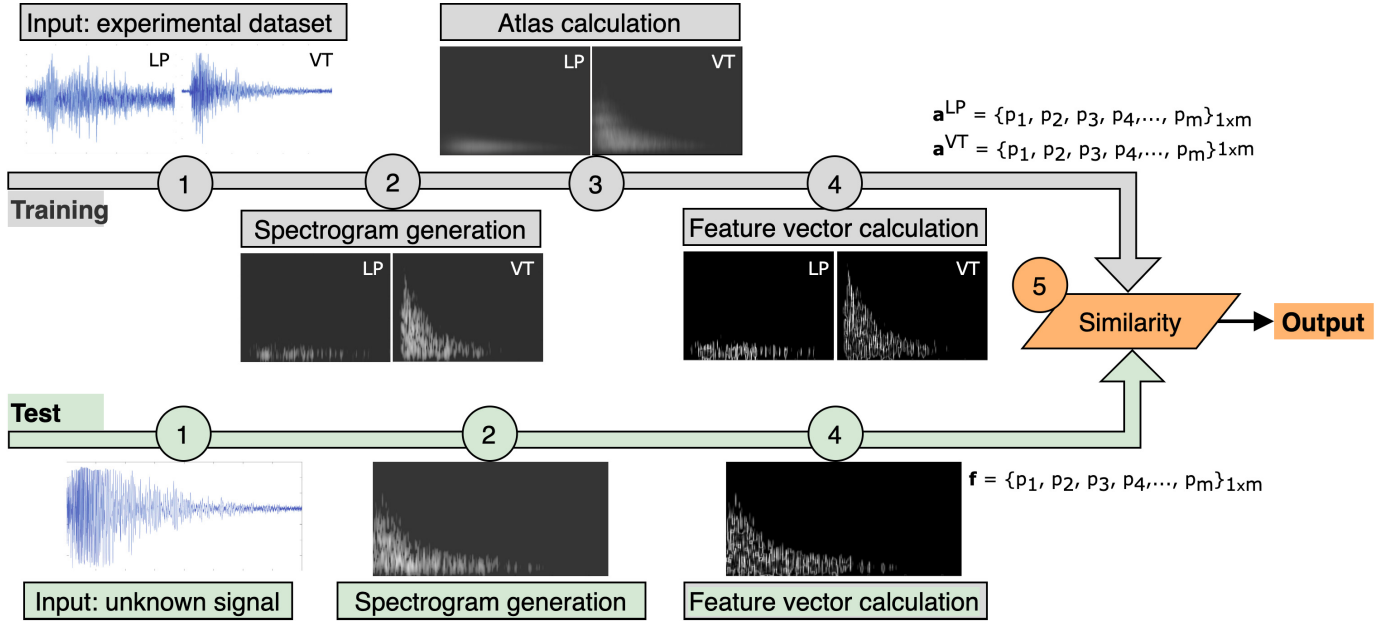


Fig. 2. Workflow of the proposed method for the training (gray color) and test (green color) steps; $\mathbf{a}$ and $\mathbf{f}$ are feature vectors used to feed the similarity-based classifier (orange color).

is based on the use of simple concepts of geometry and set theory to form morphological operators [51], which efficiently analyzes shapes and structures in binary, gray-level, and color images. Also, it is less sensitive to noise. The morphological gradient operation applied to the set of gray-level spectrogram images is defined as

$$\mathbf{G}_k = (\mathbf{I}_k \oplus \mathbf{B}) - (\mathbf{I}_k \ominus \mathbf{B}) \quad \forall \mathbf{I}_k \in \mathcal{R}^2; \quad k = 1 \dots n \quad (1)$$

where $\oplus$ and $\ominus$ denote the dilation and erosion operators of the k th spectrogram image ($\mathbf{I}_k$) using a structuring element $\mathbf{B}$, respectively; n is the total number of spectrogram images, and $\mathbf{B}$ is a kernel whose element values are formed by a combination of lines taken at 0° , 45° , 90° , and 135° (anti-clockwise) and with length 7, as defined in (2). We used the kernel with this configuration instead of classic shapes, such as disk, triangle, diamond, and rectangle. We intended with this to maximize the information about the pattern contours of the event by searching for more specific or little visible details, e.g., an attenuated part of the event's signal (see step 4 in Fig. 2)

$$\mathbf{B} = \begin{bmatrix} \begin{matrix} 135^\circ \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{matrix} & \begin{matrix} 90^\circ \\ 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{matrix} & \begin{matrix} 45^\circ \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{matrix} & \begin{matrix} 0^\circ \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{matrix} \end{bmatrix} \quad (2)$$

The subtraction of both convolutional operations (dilation and erosion) in (1) correctly estimates the edges map of the pattern in the gray-level spectrogram image. Thus, our feature vector space is composed of a set of feature vectors obtained by transforming each edges map image $\mathbf{G}_k \in \mathcal{R}^2$ into a row vector $\mathbf{f}_k \in \mathcal{R}$ (see step 4 in Fig. 2).

4) *Similarity Criteria*: The similarity is a subjective term that gives meaning to a set of observed features. By analyzing a pair of observations using an abstracted similarity function between a query image and images in the database based on some feature set [35], it is possible to gather information about how similar (or dissimilar) the observations are. This function can be used to represent different semantic contents, e.g., in image retrieval, the images are ranked according to the similarity score, where close images (lower score) are similar regarding use and purpose. A common similarity function between two feature vectors, $\mathbf{v}_q$ and $\mathbf{v}_d$, is given by

$$s(\mathbf{v}_q, \mathbf{v}_d) = d(\mathbf{v}_q, \mathbf{v}_d) \quad (3)$$

where $d(\cdot, \cdot)$ is considered a distance function on the feature space V .

Similarity has been used in some suitable feature spaces, such as global features, which commonly described the image content by color [52], shape [53], and texture [54] into a fixed-size vector. Also, similarity has been employed in feature spaces that assumed to be metric, e.g., the stimulus (a probabilistic concept in psychology) represented as points in space and color histograms of images, among others [35]. However, assuming a space to be metric implies accepting or refusing related distance axioms [34].

To better understand the theoretical description of the concept of similarity, we consider the situation under analysis, which is a problem of binary classification of LP and VT seismic events. This problem is based on a vector space, and therefore, $d(\cdot, \cdot)$ in (3) has been modified to be a distance metric between feature vectors, consequently.

Let $\mathbf{a}^{\text{LP}}$, $\mathbf{a}^{\text{VT}}$, and $\mathbf{f}$ be feature vectors in the form $\{p_1, p_2, p_3, \dots, p_m\}_{1 \times m}$ with size m and p_m the m th pixel value, where $\mathbf{a}^{\text{LP}}$ and $\mathbf{a}^{\text{VT}}$ are the computed edges map of the LP and VT atlases ($\mathbf{A}^{\text{LP}}$ and $\mathbf{A}^{\text{VT}}$), respectively, and $\mathbf{f}$ is the instance to be classified. Thus, the similarity $s(\cdot, \cdot)$ of $\mathbf{f}$ respect

to $\mathbf{a}^{\text{LP}}$ and $\mathbf{a}^{\text{VT}}$ is defined as

$$s(\mathbf{a}^{\text{LP}}, \mathbf{f}) = d(\mathbf{a}^{\text{LP}}, \mathbf{f}) \quad (4)$$

$$s(\mathbf{a}^{\text{VT}}, \mathbf{f}) = d(\mathbf{a}^{\text{VT}}, \mathbf{f}). \quad (5)$$

The interpretation of the above equations is that the closer the $s(\cdot, \cdot)$ score is to zero, the greater the probability of assigning the correct class is. Therefore, the classification of the instance $\mathbf{f}$ depends on the closeness to a specific atlas determined by

$$s^* = \arg \min_s [s(\mathbf{a}^{\text{LP}}, \mathbf{f}), s(\mathbf{a}^{\text{VT}}, \mathbf{f})] \quad (6)$$

subject to

$$C(\mathbf{f}) = \begin{cases} \text{LP}, & \text{if } s^* = s(\mathbf{a}^{\text{LP}}, \mathbf{f}) \\ \text{VT}, & \text{otherwise} \end{cases}$$

where s^* is the minimum between both computed similarity scores, and $C(\mathbf{f})$ is the predicted class of the instance $\mathbf{f}$ (see step 5 in Fig. 2).

C. Experimental Setup

This section outlines the experimental design to assess the selected distance-based classifier on two experimental data sets, containing LP and VT seismic events with and without overlapped signals of nonvolcanic origin. The training and test set creation, similarity functions configuration, validation protocol, and selection criteria are described next.

1) *Training and Test Sets*: We build disjoint training and test partitions of the data using three different training–test split scenarios 25%–75%, 50%–50%, and 75%–25%, respectively. Since the experimental data sets contain signals of both seismic events with and without overlapped signals, the partition's percentage was applied to every single event to guarantee the data stratification (representation of each class on each split), i.e., 25% of LP signals plus 25% of VT signals, and 75% of LP signals plus 75% of VT signals formed the training and test sets of the 25%–75% split scenario. This process was repeated until creating all the training and test sets. This partition methodology was previously tested in [31] with successful results.

Moreover, the test sets were used with five times stratified tenfold cross-validation method [55] before the classification stage to build 50 disjoint test folds, which enables the proposed method to learn from different input space representations. Testing on these different sets boosts the variability of the classification results of individual samples. Therefore, it will allow us to gather trustworthy information about the generalization power of the proposed method.

2) *Similarity Functions Configuration*: The proposed method implemented three different distance-based metrics, such as the L_1 -norm (Manhattan distance) defined by (7) in [36], L_2 -norm (Euclidean distance) defined by (8) in [56], and the distance correlation defined by (9), which is an improvement of Pearson's correlation to measure nonlinear association between two random vectors as well [57]. These metrics were used as the function $d(\cdot, \cdot)$ in (3) to better analyze the feature space classification from different similarity points of view. Other popular metrics, such as Minkowsky

and Mahalanobis, were not considered since the first is a generalization of the L_1 - and L_2 - norms, and the second internally computes the covariance matrix of the whole feature vector space. For our problem, this means a heavy load of information to the main memory, which is against our goal.

Regardless, the selected metrics were suited to be used in the vector space related to the problem under analysis and are defined as

$$d(\mathbf{a}, \mathbf{f}) = \sum_{i=1}^m |\mathbf{a}_i - \mathbf{f}_i| \quad (7)$$

$$d(\mathbf{a}, \mathbf{f}) = \sqrt{\sum_{i=1}^m (\mathbf{a}_i - \mathbf{f}_i)^2}. \quad (8)$$

The original distance correlation equation was adapted to analyze the complement of the scores obtained and, therefore, to satisfy the classification criteria established by (6). The modification does not alter the math sense, but it does to the interpretability. Currently, a higher value means a lower correlation between features vectors. Thereby, the employed distance is given by

$$d(\mathbf{a}, \mathbf{f}) = 1 - \frac{\text{dcov}(\mathbf{a}, \mathbf{f})}{\sqrt{\text{dvar}(\mathbf{a}) \text{dvar}(\mathbf{f})}} \quad (9)$$

subject to

$$\begin{aligned} \text{dcov}(\mathbf{a}, \mathbf{f}) &= \sum_{i=1}^m (\mathbf{a}_i - \bar{\mathbf{a}})(\mathbf{f}_i - \bar{\mathbf{f}}) \\ \text{dvar}(\mathbf{a}) &= \sum_{i=1}^m (\mathbf{a}_i - \bar{\mathbf{a}})^2 \\ \text{dvar}(\mathbf{f}) &= \sum_{i=1}^m (\mathbf{f}_i - \bar{\mathbf{f}})^2 \end{aligned}$$

where $\mathbf{a}$ and $\mathbf{f}$ are feature vectors [parameters in (4) and (5)] with size m of the event atlas and the instance $\mathbf{f}$ under analysis. $\bar{\mathbf{a}}$ and $\bar{\mathbf{f}}$ are the mean of the $\mathbf{a}$ and $\mathbf{f}$ feature vectors, and dcov and dvar are the distance covariance and distance variance, respectively.

3) *Validation Metrics*: We computed the mean of several metrics, such as ACC, AUC, precision, recall, and F1-score to assess the effectiveness of the proposed method using the three test sets (one per split) on both data sets over 50 runs. Since our method is based on similarities (distances between feature vectors), we support the discussion of results using the mean of the ACC metric. In addition, we measured the time consumed during the computation of feature vectors of seismic event atlases from the training sets (training stage) and the classification using the test sets (testing stage). The Wilcoxon statistical test [58] with a significance value of $\alpha = 0.05$ was used to assess the importance of the differences between classification schemes.

4) *Selection Criteria*: Since we explored three distance based-functions as $d(\cdot, \cdot)$ in (3) with different mathematical formulations, a golden rule was set for selecting the best classification scheme. Thus, we follow, first, the metric that provides the higher mean of ACC performance statistically

TABLE I
SUMMARY OF CLASSIFICATION PERFORMANCE ON THE D1 EXPERIMENTAL DATA SET

Similarity metric	25%-75%						50%-50%						75%-25%					
	ACC*	AUC	PRE*	REC*	F1	Wil.	ACC*	AUC	PRE*	REC*	F1	Wil.	ACC*	AUC	PRE*	REC*	F1	Wil.
L_1 -norm	88.23	0.983	87.99	92.05	0.93	$p < 0.05$	87.45	0.979	88.04	97.06	0.92	$p < 0.05$	83.31	0.976	87.41	98.18	0.92	$p < 0.05$
L_2 -norm	93.34	0.981	92.29	91.61	0.92	-	88.78	0.975	91.60	87.22	0.89	$p < 0.05$	83.68	0.964	89.73	87.26	0.88	$p < 0.05$
Dist. correlation	90.02	0.717	96.07	82.83	0.89	$p < 0.05$	85.10	0.714	96.77	76.52	0.85	$p < 0.05$	76.02	0.691	98.18	69.10	0.81	$p < 0.05$

*values in percentage and rounded to two decimal places; AUC - area under the ROC curve; PRE - precision; REC - recall; F1 - F1 score; Wil. - Wilcoxon statistical test score against the pivot; underline score means the highest ACC value (i.e., the statistical comparison pivot); bold value means the best selection

TABLE II
SUMMARY OF CLASSIFICATION PERFORMANCE ON THE D2 EXPERIMENTAL DATA SET

Similarity metric	25%-75%						50%-50%						75%-25%					
	ACC*	AUC	PRE*	REC*	F1	Wil.	ACC*	AUC	PRE*	REC*	F1	Wil.	ACC*	AUC	PRE*	REC*	F1	Wil.
L_1 -norm	91.30	0.997	90.94	100	0.95	$p < 0.05$	89.66	0.996	91.14	100	0.95	$p < 0.05$	86.53	0.992	90.08	100	0.95	$p < 0.05$
L_2 -norm	96.88	0.997	96.52	100	0.98	$p < 0.05$	95.15	0.996	97.72	100	0.99	$p < 0.05$	89.43	0.992	97.92	97.72	0.98	$p < 0.05$
Dist. correlation	<u>97.32</u>	0.757	100	99.20	0.99	-	95.44	0.753	100	97.13	0.98	$p < 0.05$	88.99	0.743	100	94.55	0.97	$p < 0.05$

*values in percentage and rounded to two decimal places; AUC - area under the ROC curve; PRE - precision; REC - recall; F1 - F1 score Wil. - Wilcoxon statistical test score against the pivot; underline score means the highest ACC value (i.e., the statistical comparison pivot); bold value means the best selection

at $p < 0.05$, and second, if there was a tie performance in the mean of ACC scores, the metric with the simplest mathematical formulation is preferred to mitigate the algorithmic complexity of the method.

We used the t-Distributed Stochastic Neighbor Embedding (t-SNE) technique [59] to visualize the multidimensional space of both experimental data sets into a bidimensional one. Also, for supporting the comparison between the original and computed feature spaces, the t-SNE was always applied after the classification stages to avoid transforming the data before feeding the similarity functions.

The implementation and application of the proposed method and the t-SNE technique were made in MATLAB (Release R2019a) [60] on an Intel Xeon processor-based workstation with 2.3 GHz and 128-GB RAM.

IV. RESULTS AND DISCUSSION

According to the experimental setup section, the proposed method was validated using three different similarity functions on two experimental data sets (with and without overlapped signals) with three different training-test split scenarios. The obtained results are described next, highlighting the satisfactory performance based on the mean of the ACC metric.

A. Performance on the Experimental Data Sets

In the D1 data set, except for the distance correlation metric in the 75%-25% split, which obtained the worst ACC mean value of 76.02%, the remaining metrics reached ACC mean scores over the 83%, which is considered a reasonable performance. However, the best ACC mean-based classification performance of 93.34% was reached by the L_2 -norm metric using the 25%-75% split. This result was statistically superior ($p < 0.05$) compared with the rest, as shown in Table I.

Concerning the created split scenarios, all metrics performed better in the 25%-75%, 50%-50%, and, finally, 75%-25% scenarios. These results were very interesting. Almost always, traditional MLCs demand a certain amount of samples for successful model training without overfitting. In contrast,

the similarity metrics preferred training the model (computing the seismic event atlases) with a small number of samples, and that was sufficient enough to provide good similarity scores. This situation could be supported by the fact of being the D1 data set, a real-life data set, containing seismic events with overlapped signals. These specific samples introduce more variations when creating the seismic event atlas that could affect the classification performance (higher similarity scores). Thus, the probability of taking random samples with overlapped signals in the 25%-75% split is lower than compared with the other scenarios, which attained inferior classification performances.

Regarding the similarity metrics as classifiers, the L_2 -norm reached the best classification scores for all scenarios in this data set. These results were expected since the L_2 -norm is the Euclidean distance, and by definition, it finds the shortest path (small length) between two points (feature vectors) in the space. Meanwhile, the L_1 -norm (Manhattan distance) determines the distance between two points using a grid layout mechanism in the space (definition constrain); thus, it produces lower scores with respect to the L_2 -norm metric. Finally, the distance correlation computes the proximity of two points using their correlation value. As seismic events with overlapped signals introduce more variation in the computed atlas, they become less correlated with the individual samples. This explains the slightly lower results obtained by this metric.

On the other hand, all metrics reached good classification results in the D2 data set. The obtained mean of ACC scores was higher than 86%, which is considered a satisfactory performance. The higher score was obtained by the distance correlation metric, reaching a mean ACC score of 97.32% in the 25%-75% split scenario. However, this result did not represent a statistical difference to 96.88% reached by the L_2 -norm metric, as it could be seen in Table II.

In this data set, all the split scenarios provided good classification performances on all metrics. Since the D2 data set does not contain samples with overlapped signals, there are not higher variations in the computed seismic event atlas with respect to the individual samples. Thus, the number of

TABLE III
SUMMARY OF THE TIME CONSUMED DURING THE TRAINING AND TEST STAGES ON EACH OF THE
THREE SPLIT SCENARIOS OF BOTH EXPERIMENTAL DATA SETS

	D1 dataset						D2 dataset					
	25%-75%		50%-50%		75%-25%		25%-75%		50%-50%		75%-25%	
	TT (s)	TS (s)	TT (s)	TS (s)	TT (s)	TS (s)	TT (s)	TS (s)	TT (s)	TS (s)	TT (s)	TS (s)
Training stage												
A^{LP} and A^{VT} calculation	0.292	0.002	0.376	0.002	0.388	0.001	0.269	0.002	0.351	0.001	0.371	0.001
Test stage												
L_1 -norm	5.693	0.011	3.754	0.011	1.643	0.010	4.630	0.011	2.892	0.011	1.236	0.009
L_2 -norm	6.097	0.012	4.031	0.012	1.798	0.011	4.870	0.012	3.053	0.011	1.328	0.010
Dist. correlation	7.857	0.016	4.948	0.015	2.538	0.015	6.673	0.016	4.013	0.015	1.938	0.014

TT - total time consumed; TS - time consumed per sample; all values were rounded to three decimal places

instances used to train (i.e., to compute the atlas) the similarity metrics is no longer an essential aspect to consider. In such a sense, there are no lousy classification performances in this data set, independently of the employed training–test split scenario.

Concerning the similarity metrics as classifiers in the D2 data set, the distance correlation and L_2 -norm reached the best performance without significant ($p < 0.05$) score difference between them. Both metrics consistently registered successful classification performance across all scenarios in this data set. On the contrary, the L_1 -norm metric performance was lower. The good classification results of the distance correlation metric in this data set can be explained by the absence of seismic events with overlapped signals. This makes the computed event atlas highly correlated with the individual samples. Besides, from Table II, it should be noted that the distance correlation, and the L_2 -norm metrics reached a similar mean of ACC score per split scenario. The small difference values of 0.44 (25%–75% split), 0.29 (50%–50% split), and 0.44 (75%–25% split) demonstrated that using more instances to compute the seismic event atlas (model training) brings them more closely to its class centroid. Thus, both metrics were particularly benefited from the D2 data set configuration. This explains the small classification difference between them.

Overall, the ACC-based classification performance slightly decreased from the 25%–75% split to the 75%–25% split for all similarity functions on both data sets (see Tables I and II). These results were expected and are linked to the number of samples used to train the models. As long as more samples are used to form the seismic atlases, more variations are introduced to the seismic pattern itself due to their computation (an average), which affects the performance in these classifiers (based on similarity). This explains why the classification score of test sets in the later splits is lower with respect to the first one.

In terms of execution time, the total time (TT) and the time consumed per-sample (TS) did not exceed 0.388 and 0.002 s across all split scenarios of both data sets, respectively. These scores represented fast execution results and are summarized in Table III. From this table, it is possible to read that, during the test stage, the L_1 -norm metric performed faster than the remaining metrics, reaching TT and TS scores between the best (75% – 25% split) and worst (25% – 75% split) scenarios in the range from 1.236 to 5.693 and 0.009 to 0.011

s, respectively. These results did not represent a big difference against those obtained by the L_2 -norm metric, which reached TT and TS scores between the best and worst scenarios in the range from 1.328 to 6.097 and 0.010 to 0.012 s, respectively. The small-time difference of no more than 0.001 s between both metrics during the classification stage could be considered imperceptible. On the other hand, the distance correlation metric needed a bit more time to fulfill the task. This situation is explained by the mathematical formulation of the employed metrics, which directly affected the execution time.

Moreover, it should be noticed that all metrics consumed more time in the D1 versus the D2 data set. This result was expected since the D2 data set does not contain samples with overlapped signals. Thus, a lesser number of samples ideally decreased the execution time of all metrics.

According to the selection criteria, the L_2 -norm is the best similarity function to be used as the classifier by the proposed method for classifying LP and VT seismic events. It was statistically similar to the distance correlation in terms of classification performance on both types of data sets but with the simplest mathematical formulation. In accordance with the definition of real time described in [5], the proposed method received, processed, and returned the classification answer quickly enough to be considered as produced in real time.

B. Performance on the Original Versus Computed Feature Space

We carried out the comparison between both feature spaces based on the highest classification score obtained by each similarity function in both experimental data sets, as summarized in Table IV. From this table, it is possible to interpret that, except for the L_1 -norm, which reached the highest score in the original feature space (i.e., gray-level spectrogram space), the remaining metrics provided better classification performances on the computed edges map feature space. These results were partially expected and related to the data dispersion in the feature spaces. Since the L_1 -norm metric measures the distance between two points considering a grid layout search strategy, scattered feature spaces are then beneficial to it. In contrast, the remaining metrics advocate for more clustered feature spaces as the computed edges map. Fig. 3 shows a visualization of the data dispersion of both experimental data sets and the two feature spaces using the t-SNE technique.

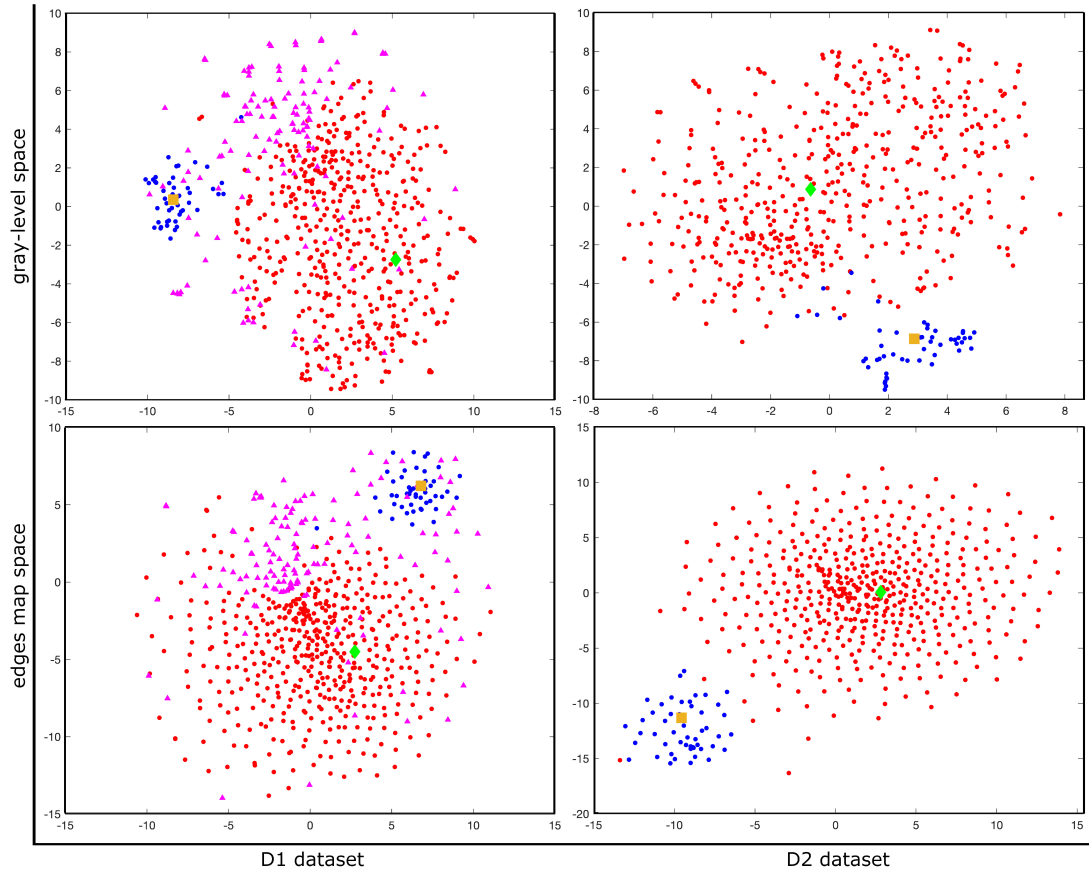


Fig. 3. Data visualization of the D1 (left column) and D2 (right column) data sets using the t-SNE technique. The LP (red circles), VT (blue circles) seismic events, LP atlas (green diamonds), VT atlas (mustard squares), and samples (LP and VT) with overlapped signal (magenta triangles) in the original gray-level (top row) and edges map (bottom row) feature spaces, respectively.

TABLE IV

ACC-BASED CLASSIFICATION PERFORMANCE COMPARISON PER SIMILARITY FUNCTION BETWEEN BOTH FEATURE SPACES

Similarity metric	Original feature space		Computed feature space	
	D1 dataset	D2 dataset	D1 dataset	D2 dataset
L_1 -norm	92.78	95.26	88.23	91.30
L_2 -norm	92.50	96.39	93.34	96.88
Dist. correlation	87.43	95.21	90.02	97.32

All values are in percentage and were rounded to two decimal places; bold value means the best performance between datasets of the feature spaces

From this figure, it is possible to see that, in the original feature space [see Fig. 3 (top row)], samples of the same class are separated from each other and with respect to their computed seismic event atlas. This dispersion makes the classification task more difficult. For example, in the D1 data set, the majority of samples with overlapped signals (magenta triangles) seen to be more close to the VT (mustard square) than the LP (green diamond) atlas, which is a wrong assumption because there are samples of both types of seismic events. Likewise, in the D2 data set, there are some samples of VT class (blue circles) that is wrongly closer to the LP atlas, and vice versa. Thus, such samples would be misclassified.

On the other hand, in the computed edges map feature space [see Fig. 3 (bottom row)], samples are more clustered to their event atlas. For example, in the D1 data set, the instances with

overlapped signals (magenta triangles) are better-distributed and closer to their event atlas (opposite effect to the original space). Similarly, in the D2 data set, samples were regrouped around their atlas. For instance, in VT samples (blue circles) that would be wrongly classified (false positive) in the D1 data set, there are located near to their atlas and, thus, would be correctly classified.

Overall, the original feature space is centered in the pixel intensity value as relevant information of the seismic pattern, neglecting the redundant information (several pixels with the same value). Meanwhile, the computed feature space is centered on preserving only the seismic pattern's edge information, minimizing the redundancy effect. Thereby, the latter provided an effective environment for the classification of both types of seismic events by facilitating the clustering of the samples around the correct event atlas (i.e., centroid).

C. State-of-the-Art-Based Comparison

Although it is not possible to establish a direct statistical comparison against some of the developed state-of-the-art methods described in the literature, such as in [6], [19], [22], and [40], due to the different experimental conditions, others methods were experimented using subsets of the MicSigV1 data set, such as the proposed method. Either way, we aimed

TABLE V
ACC-BASED CLASSIFICATION PERFORMANCE REPORTED BY SOME OF THE STATE-OF-THE-ART METHODS AND THE PROPOSED METHOD

Method	Number of samples	Computed features	Spectrogram images	Overlapped signals	ACC*	Time complexity [‡]
ANN [19]	1033	8	no	no	94	$\mathcal{O}(n^4)$
HMM [40]	512	39	no	no	90	$\mathcal{O}(n^2)$
SVM [6]	105000	102	yes	no	92	$\mathcal{O}(n^3)$
CNN [22]	15895	-	yes	no	97	$\mathcal{O}(n^4)$
SPCC+KNN [37]	514	-	yes	no	95	$\mathcal{O}(n)$
ANN [39]	914 ⁺	6	no	no	97	$\mathcal{O}(n^4)$
DT [39]	914 ⁺	3	no	no	96	$\mathcal{O}(n)$
ANN [13]	637 ⁺	17	no	no	95	$\mathcal{O}(n^4)$
RF [13]	637 ⁺	17	no	no	93	$\mathcal{O}(n)$
linear SVM [25]	914 ⁺	5	no	no	97	$\mathcal{O}(n^3)$
GMM [31]	667 ⁺	2	no	no	94	$\mathcal{O}(n^2)$
DCNN1 [20]	1108 ⁺	-	yes	yes	95	$\mathcal{O}(n^4)$
Proposed method	668⁺	-	yes	yes	97	$\mathcal{O}(n)$

ACC - accuracy; n - size of the training dataset; *values in percentage and rounded to the closest integer; ⁺ samples subset of the *MicSigV1* dataset; [‡] worst case time complexity estimation according to [61]–[67]

to carry out the comparison based on the ACC scores reported by these works, as shown in Table V.

From this table, it is possible to read that the majority of MLCs reached ACC scores ranging from 90% to 97%, the CNN, ANN, and linear SVM being the models that provided the highest classification scores in subsets of the *MicSigV1* data set. Observe that the proposed method reached the highest score of 97% as well, even competing with the CNN performance in [20] on a subset of the *MicSigV1* data set containing overlapped signals. Moreover, similar to CNN-based approaches, our technique offers a competitive performance without computing handcrafted features from the raw signal's traditional domains (i.e., time, frequency, and scale) but with the advantage of being far less computationally expensive.

Besides, it should be noted that all of such methods, except the CNN-based approaches, implemented a traditional data analysis workflow, involving data processing, feature selection, and classification based on a wide variety of MLCs. Also, their time complexity varies in the range from $\mathcal{O}(n)$ to $\mathcal{O}(n^4)$. In contrast, the proposed method is based on the computation of an average seismic event pattern (i.e., atlas) in the training and distances (i.e., similarity) between feature vectors in the classification for the time complexity of $\mathcal{O}(n)$. Since both actions perform in linear time (see Table V), the time execution is faster, and the algorithm complexity is less compared with the vast majority of state-of-the-art methods. These facts enable the proposed method as a practical and beneficial scheme that could be implemented in real-time seismic events' classification environments.

D. Study Limitations

There are some practical limitations of the proposed method that have to be taken into consideration for further applications.

- 1) *Retraining the Model at Some Point in Time* This action must be mandatory to keep updated the pattern atlas of considered seismic events. Since the atlas depends on the mean of the events, as the number of new incoming samples increased, the events pattern could vary as well, and the retraining process becomes crucial to avoid misclassification.

- 2) *Signal Dependent Model* The model works accurately when it is well trained with data recorded at the same station (equal data acquisition parameters). Therefore, any subsequent deployment to another station must include retraining the model to function properly in the new environment.
- 3) *Sensitive to Signal Overlapping* The model performed better when using the D2 data set (ideal), which does not contain samples with overlapped signals. Since the computed feature space is a vector space, the feature vectors do not take care of the overlapped signal location. Thus, it becomes a part of the relevant information that will affect both the training, when computing the seismic event atlases, and the test, when determining the similarity scores. Under this situation, a signal preprocessing step will be needed to remove the unnecessary information before feeding the proposed method.
- 4) *Memory Load-Dependent on Number of Event Classes* The atlas of each event class remains in the main memory to classify new incoming samples. As long as the number of classes to be analyzed increases, the memory loading (for hosting all the computed atlases) and execution time (during the classification) could increase as well.

V. CONCLUSION

In this work, we propose a new approach based on a suit combination of mathematical morphology and similarity criteria techniques to classify LP and VT seismic events applied to the Cotopaxi volcano. The proposed method computes a new feature space based on the edges map of the seismic events pattern represented in their corresponding gray-level spectrogram images, which is used to feed a set of distance metric-based classifiers. The L_2 -norm was selected as the best metric to be implemented by the proposed method. In terms of classification performance, it was statistically superior in the D1 data set (seismic events with overlapped signals of non-volcanic origin) and similar in the D2 data set (events without overlapping signals) with respect to the other metrics, reaching ACC mean scores of 93.34% and 96.88%, respectively. These results demonstrated that the computed edges map feature space is a better environment for separating both seismic event

classes compared with the original gray-level space. Regarding the execution time, the proposed method required a total and per-sample times of no more than 0.388 and 0.002 s during the training stage and a per-sample time of no more than 0.012 s in the testing stage. Finally, the proposed method execution is faster, and the algorithm complexity is less compared with the state-of-the-art methods, which makes it a practical and beneficial scheme to implement in a real-time seismic events' classification environment.

As future work, we plan to implement two major improvements regarding the study limitation. First, we will consider the bidimensional edges map feature space to experiment with the use of template-based matching techniques as classifiers. With this, it is possible to segment the overlapped signals in the feature space and, thus, before entering the classification stage. Second, the inclusion of samples from different observatory stations to improve the versatility of the proposed method. In addition, the incorporation of other types of events, such as tremors, icequakes, very-long periods, and hybrids, is also required to extend its applicability.

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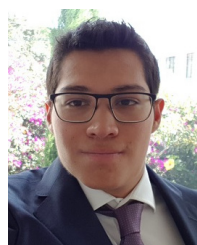
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